

Workload cost modeling approach to choose table format on IBM Warehouses

The challenge

Warehouses are central to Enterprise Data Processing needs. Data has become more Open in nature and does not often reside within Warehouses anymore. This makes it key for Warehousing to not only optimally process data that resides in a warehouse, but at the same time integrate and easily access Open File and Open Table Formats, not compromising on performance aspects.

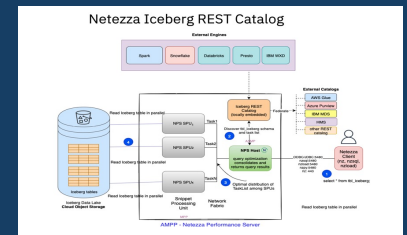
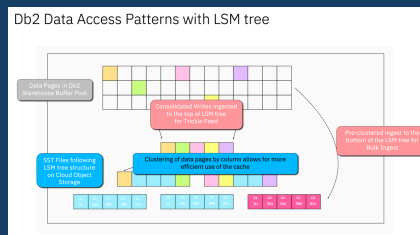
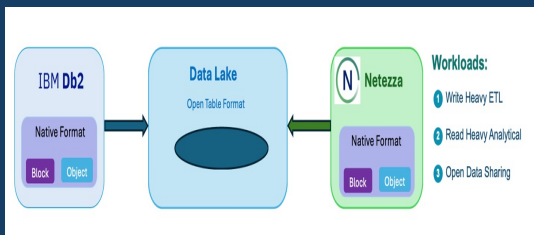
Performance Considerations

The speed at which table metadata can be retrieved and utilized governs performance of a query execution where metadata processing time heavily depends on the type of table format that was used to store the metadata.

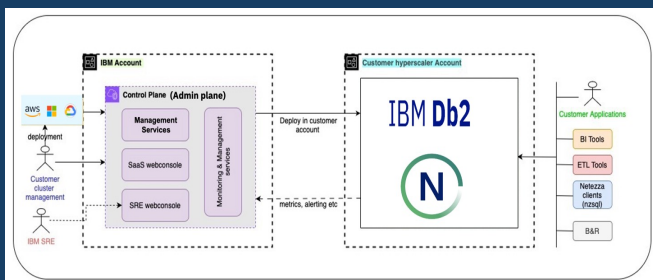
Key Objectives

- Select a table format for high performant metadata planning over a data lake table
- Take a workload cost modeling approach to choose table format for high performant query planning in a Lakehouse system
- For given table format identify best fit engine to optimize cost in Cloud deployments

Data Lakehouse Architecture



The above architecture can be deployed using recently introduced **Bring Your Own Cloud (BYOC)** deployment models for **Db2 Warehouse**, and **Netezza** enabling customers to benefit from a SaaS-like operational experience while allowing you to run your data processing directly on your own infrastructure.



IBM Warehouses Db2 and Netezza can be deployed in any customer pattern					
	Self-managed software	Cloud-managed IaaS	Cloud-managed container PaaS	Fully-managed SaaS	Bring-Your-Own-Cloud SaaS
Db2					
Db2 Warehouse					
Db2 pureScale					
Netezza					

IBM data warehouse solutions introducing capabilities that make them a favorable choice in Warehouse and Lakehouse market.